

3.1.2 Linear motion

Learning outcomes		Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>		
(a)	(i) the equations of motion for constant acceleration in a straight line, including motion of bodies falling in a uniform gravitational field without air resistance $v = u + at \qquad s = \frac{1}{2}(u + v)t$ $s = ut + \frac{1}{2}at^2 \qquad v^2 = u^2 + 2as$	M2.2, M2.4, M3.3 HSW9
	(ii) techniques and procedures used to investigate the motion and collisions of objects	PAG1 Apparatus may include trolleys, air-track gliders, ticker timers, light gates, data-loggers and video techniques. HSW4, 9, 10
(b)	(i) acceleration g of free fall	
	(ii) techniques and procedures used to determine the acceleration of free fall, to include using trapdoor and electromagnet arrangement and light gates and timer arrangement	PAG1 HSW4, 5, 7 Determining g in the laboratory.
(c)	reaction time and thinking distance; braking distance and stopping distance for a vehicle.	HSW5, 9, 10, 11, 12